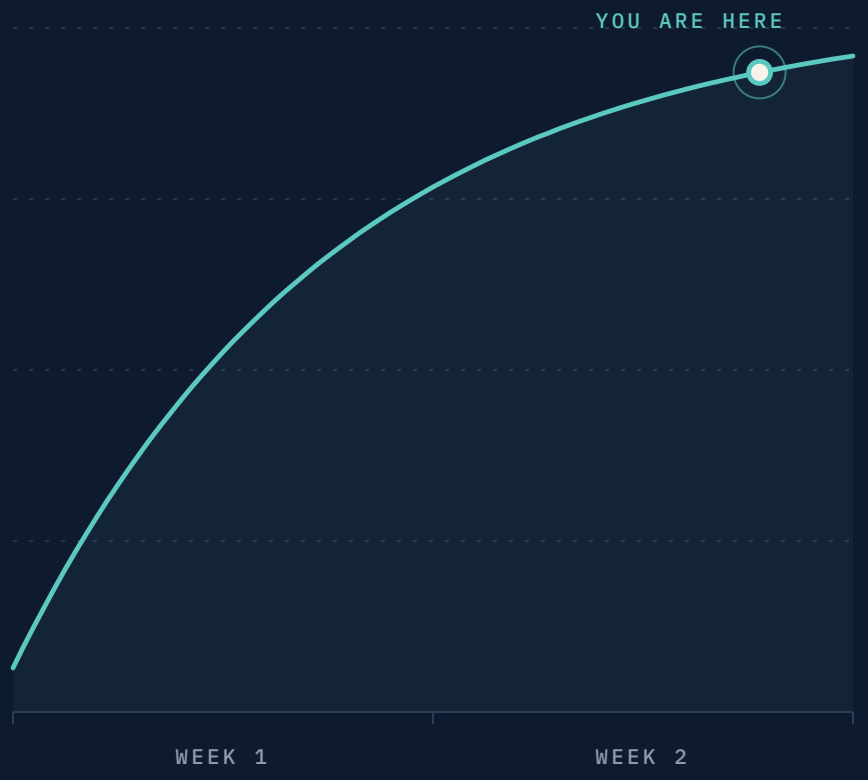
WEEK 2 · DAY 4 · OVERVIEW

Two Series

Cointegration and Mean Reversion



Created by **Jeff Lenamon**

ABOUT 45 MIN WITH EXERCISES / 10 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Statement

BUSINESS CONTEXT

Day 3 found that two wandering series nearly always find a line between them, and promised a test for a link that survives.

A desk follows pairs of series: the 10 Yr against the 5 Yr, both real, and the HY spread against the IG spread, both synthetic.

The 2s10s slope is already a gap between two series with the ratio fixed at 1, and Day 3 found that it wanders.

Today the data picks the ratio, a test says whether the gap holds a level, and a half-life says how fast it closed.

Every number here describes 2015 to 2022, and none of it is a trade, an entry, or a view on rates or spreads.

OBJECTIVES

Compare a pair's correlation in levels and in changes, fit a hedge ratio by eye and by least squares, build the gap, and map where a pair moved from each place it stood.

Run the Engle-Granger test with **`coint`** and read its statistic and p-value, measure a half-life from an AR(1) on the gap and check its sign, and follow the hedge ratio through the years and onto later dates.

ABOUT 45 MIN WITH EXERCISES / 10 SECTIONS / 4 EXERCISES

Your Route Through Day 4

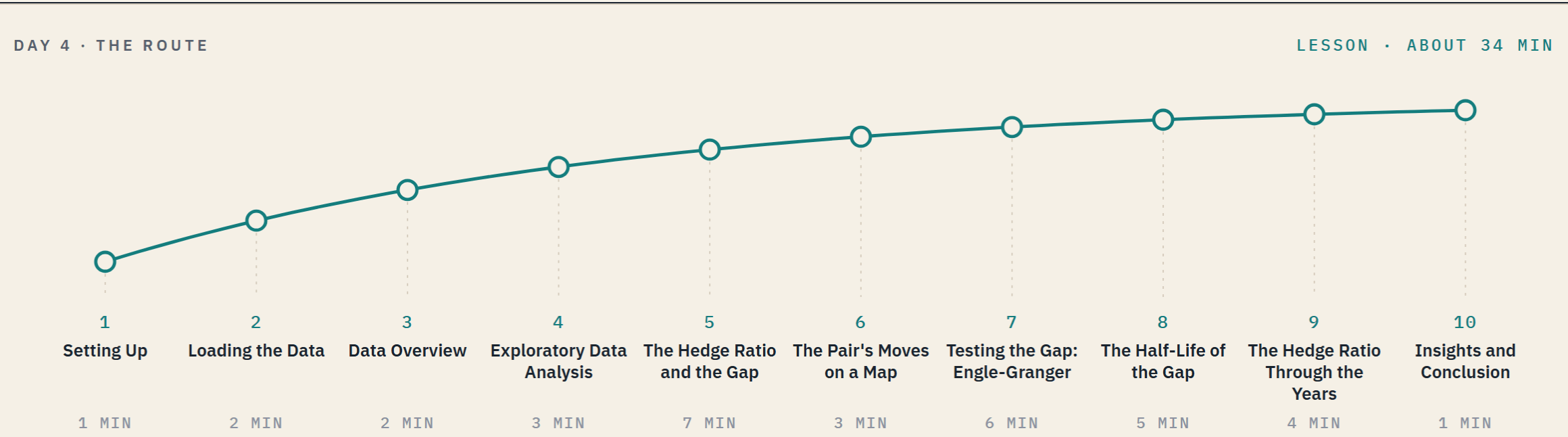


FIGURE 1 Ten sections, in order, with about how long each takes.

Time. About 34 minutes for the lesson, then about 12 for the four exercises.

Fix the seed and load the libraries.

Setting Up

THE SETUP

```
import os
from pathlib import Path
SEED = 42
DATA = "../../../data/"
if not Path(DATA + "benchmark.csv").exists():
    DATA = "https://raw.githubusercontent.com/lenamonj/python-fixed-income/main/data/"
```

THE LIBRARIES

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
from statsmodels.tsa.stattools import adfuller, coint
from sklearn.model_selection import TimeSeriesSplit
```

Read both files and build four series.

Loading the Data

READING THE TWO FILES

```
yields = pd.read_csv(DATA + "treasury_par_yields.csv", parse_dates=["date"], index_col="date")
spreads = pd.read_csv(DATA + "course3_synthetic_spreads.csv", parse_dates=["date"], index_col="date")
CUT = "2022-12-30"
development = yields.loc[:CUT]
```

FOUR SERIES IN BP

```
levels = pd.DataFrame({"5 Yr": (development["5 Yr"] * 100).round(2),
                       "10 Yr": (development["10 Yr"] * 100).round(2),
                       "IG": spreads.loc[:CUT, "ig_spread_bp"],
                       "HY": spreads.loc[:CUT, "hy_spread_bp"]})
changes = levels.diff().dropna()
print(levels.shape, changes.shape, levels.index[0].date(), "to", levels.index[-1].date())
```

```
(2001, 4) (2000, 4) 2015-01-02 to 2022-12-30
```

Data Overview

THE FOUR LEVELS, IN BP

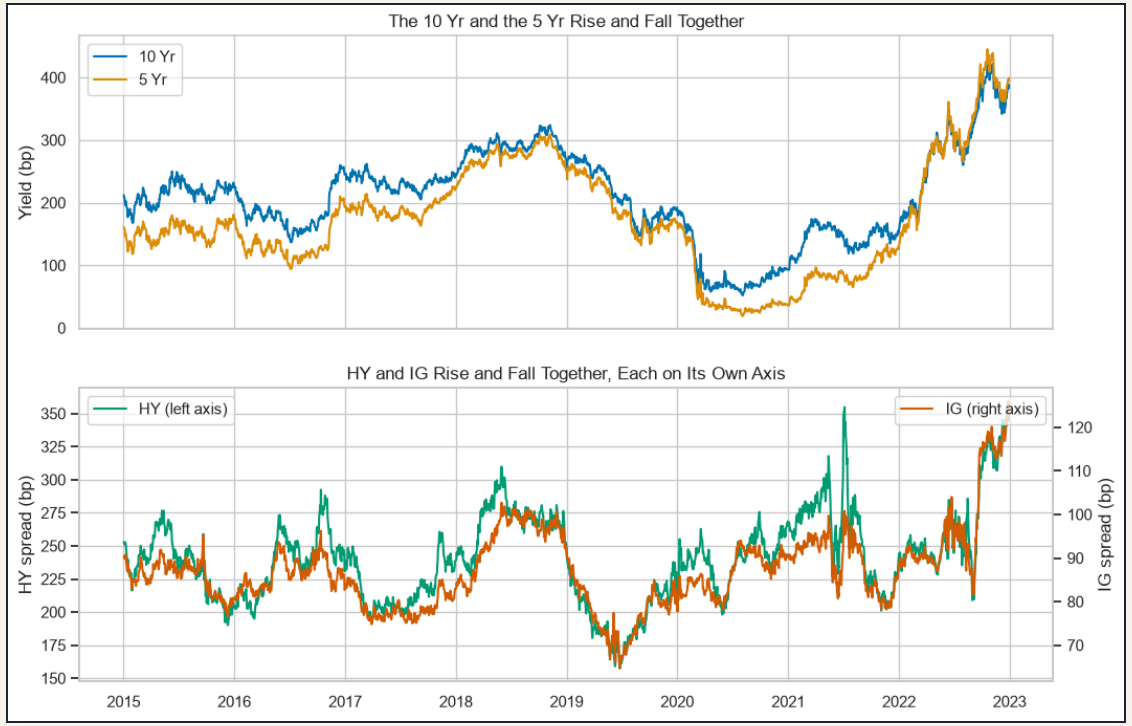
	5 YR	10 YR	IG	HY
count	2001.0000	2001.0000	2001.0000	2001.0000
mean	173.1474	207.8826	87.1356	240.5106
std	88.7875	73.8736	8.9974	33.1905
min	19.0000	52.0000	64.7000	157.5000
25%	118.0000	160.0000	81.3000	215.6000
50%	165.0000	212.0000	86.4000	240.4000
75%	233.0000	254.0000	91.8000	262.1000
max	445.0000	425.0000	125.9000	359.1000

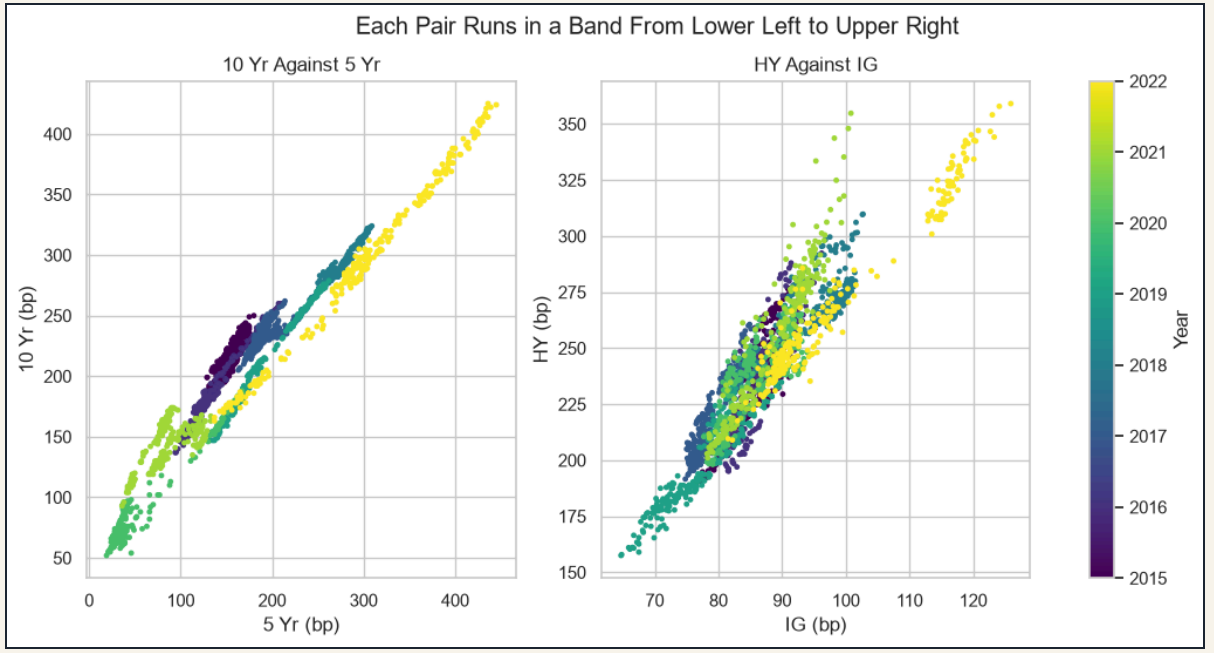
ADF'S P-VALUE ON EACH LEVEL

```
5 Yr    0.9620
10 Yr   0.8699
IG      0.5529
HY      0.0793
Name: ADF p-value, dtype: float64
```

Each level wanders on its own. Today asks whether two of them wander together.

Exploratory Data Analysis

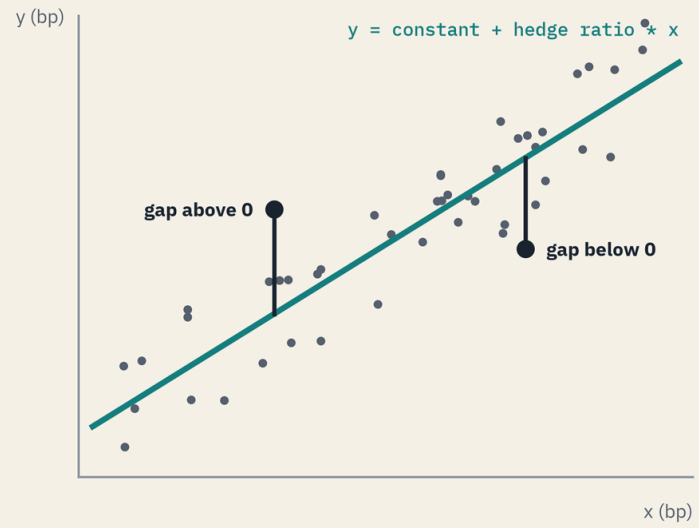




LEVELS AND CHANGES

	levels	changes
10 Yr on 5 Yr	0.9741	0.9316
HY on IG	0.9222	0.8590

The Hedge Ratio and the Gap




```
gap = y - constant - hedge ratio * x
```

the hedge ratio: bp of y per bp of x

the gap: how far y sat from the line, in bp

```
2s10s = 10 Yr - 1 * 2 Yr
```

a gap with the ratio fixed at 1 and a constant of 0

FIGURE 2 y against x with its least-squares line: the gap is how far y sat from the line, in bp; 2s10s fixes the ratio at 1.

THE QUESTION

Q. Which hedge ratio of HY on IG gives the gap its smallest standard deviation? Write down a number.

THE GAP AT ANY RATIO

```
def hy_gap(ratio):
    gap = levels["HY"] - ratio * levels["IG"]
    return gap - gap.mean()
```

THE HEDGE RATIO

The **hedge ratio** is the bp y moved per bp of x, a number fitted on this file's dates, and the **gap** is y less a constant less the ratio times x, in bp.

Both gaps average zero by construction. Whether a gap keeps coming back to zero is the question.

The Pair's Moves on a Map



THE ARROWS

`ax.quiver(x, y, dx, dy, angles="xy", scale_units="xy", scale=1)` draws one arrow per cell from (x, y), its length in the chart's own bp.

THE MAP

A **vector field** draws an arrow at each spot of a map: here, the average move a pair made from that spot.

CELLS THAT MOVED TOWARD ZERO, AND THE CORRELATION OF THE GAP WITH ITS MOVE

```
10 Yr on 5 Yr: 21 of 31 cells moved toward 0 | correlation of the gap and its move: cells -0.42, dates -0.20
HY on IG: 16 of 22 cells moved toward 0 | correlation of the gap and its move: cells -0.78, dates -0.42
```

One pair leaned back to its line, the other slid along it. A test turns the picture into a number.

Testing the Gap: Engle-Granger

THE QUESTION

Q. Draw 1,000 pairs of independent random walks, fit each pair's line by least squares, and run Day 3's ADF on each fitted gap. How many gaps will it clear of a unit root at 5 percent? Write down a number.

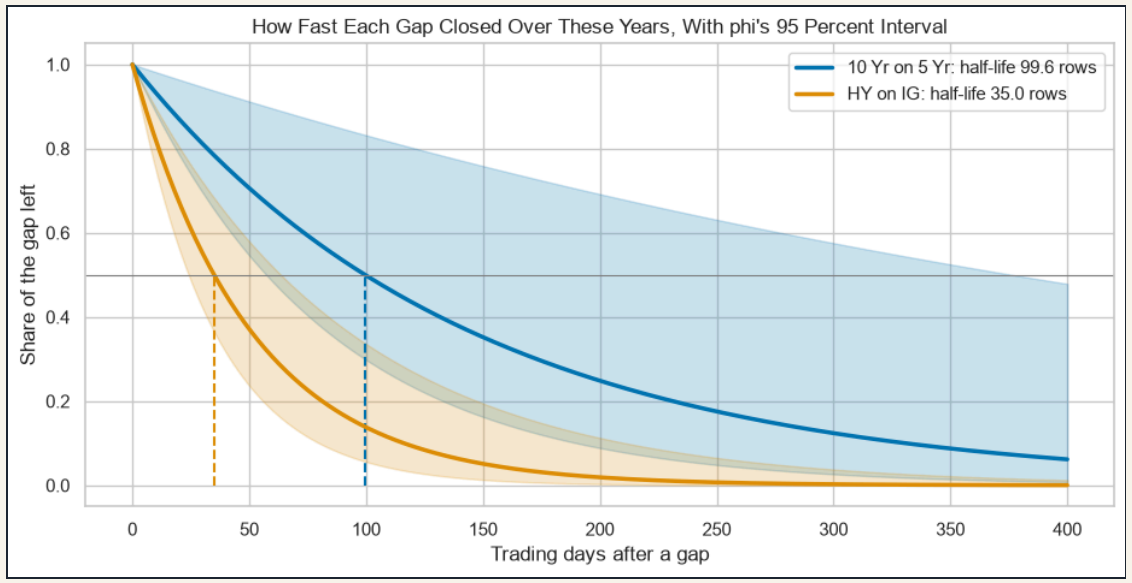
THE TWO STEPS

The **Engle-Granger test** takes two steps: fit the line by least squares, then test the gap for a unit root.

A THOUSAND PAIRS OF UNRELATED WALKS

```
rng = np.random.default_rng(SEED)
walk_pairs = rng.standard_normal((1000, 2, 500)).cumsum(axis=2)
gap_tests, walk_tests = [], []
for y, x in walk_pairs:
    ratio, constant = np.polyfit(x, y, 1)
    gap_tests.append(adfuller(y - constant - ratio * x, maxlag=0, autolag=None, result_object=True))
    walk_tests.append(adfuller(y, maxlag=0, autolag=None, result_object=True))
```

The Half-Life of the Gap



PHI AND THE HALF-LIFE

```
10 Yr on 5 Yr: phi 0.9931, half-life 99.6 trading days
HY on IG: phi 0.9804, half-life 35.0 trading days
```

THE QUESTION

Q. Fit the same AR(1) to 1,000 random walks of 2,001 rows each. How many will give a **`phi`** below 1, and so a half-life? Write down a number.

PHI'S 95 PERCENT INTERVAL, AND THE HALF-LIFE AT EACH END

```
10 Yr on 5 Yr: phi 0.9880 to 0.9982, half-life 57.3 to 376.4 trading days
HY on IG: phi 0.9716 to 0.9892, half-life 24.1 to 63.5 trading days
```

HY on IG: 24.1 to 63.5 trading days around 35.0, the top end more than twice the bottom.

The Hedge Ratio Through the Years

THE QUESTION

Q. Take fold 1's training line for the 10 Yr on the 5 Yr, and build the gap with it on the next 252 dates. What share of those dates will sit beyond 2 standard deviations of the training gap? Write down a percent.

THE LINE FITTED ON EACH FOLD'S TRAINING LEVELS

```
        ratio, constant = np.polyfit(x.iloc[train], y.iloc[train], 1)
```

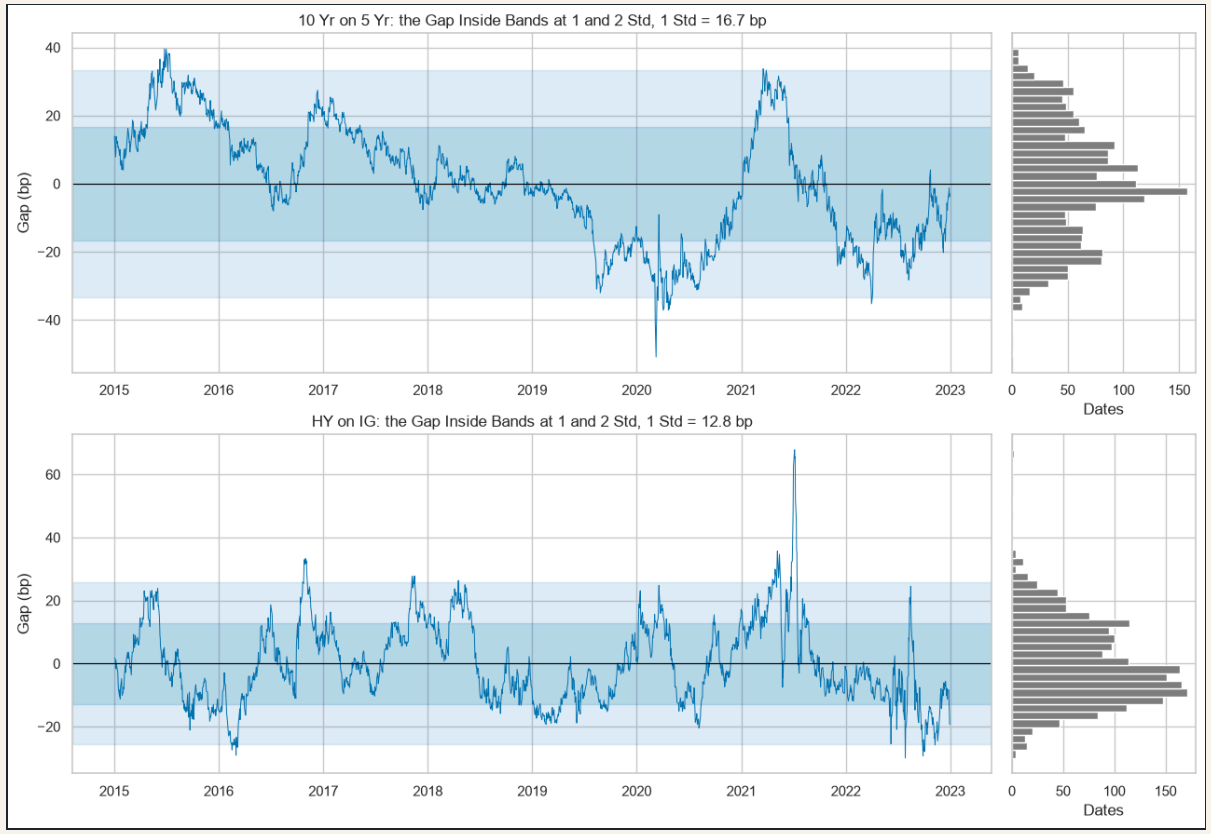
EACH DATE'S RATIO FROM THE 252 DATES UP TO IT, AND EACH FOLD'S RATIO, FOR THE 10 YR

```
    rolling_ratio[label] = y.rolling(252).cov(x) / x.rolling(252).var()
    fold_ratio[label] = [np.polyfit(x.iloc[train], y.iloc[train], 1)[0] for train, test in folds]
```

```
10 Yr on 5 Yr: rolling from 0.40 (2022-02-04) to 1.81 (2021-03-30) | folds [0.906 0.743 0.762 0.889 0.861]
```

A trading year's ratio can sit far from the ratio of all eight years.

Insights and Conclusion



LONGEST STAY ON ONE SIDE

The Treasury gap spent 462 dates in a row on one side of zero, nearly two years; the HY gap's longest run was 222.

CROSSINGS OF ZERO

The HY gap crossed zero 87 times, the Treasury gap 69.

THE RUNS AND THE BANDS

```
side = np.sign(gap)
runs = side.groupby((side != side.shift()).cumsum()).size()
beyond = (gap.abs() > 2 * gap.std()).sum()
```

Test the gap with `coint` first, then measure its half-life, then check its line on later dates.

You Can Now

- Compare a pair's correlation in levels and in changes, and say why it cannot settle a link.
- Fit a hedge ratio by least squares and build the gap.
- Draw a vector field of a pair's moves from where it stood.
- Run Engle-Granger with **`coint`** and read its statistic and p-value.
- Measure a gap's half-life from an AR(1), with its interval, and check its sign.
- Follow a hedge ratio through the years and test a training line on later dates.

NEXT
WEEK 2 · DAY 5

Course 5B Capstone

The capstone brings the course's models together and opens the held-out years once.